

CSV201:SOURCE CODE MANAGEMENT

L:3 T:0 P:0 Credits:3

Course Outcomes: Through this course students should be able to

- CO1 :: Understand Devops and Software Delivery concepts
- CO2 :: Describe the Source Code Management and its benefits
- CO3 :: Study of Source Code Management Architecture and its types
- CO4 :: Understand different phases of Git and ability to transfer files into different phases
- CO5 :: Understand the managing the work with Git
- CO6 :: Understand how to Leveraging GitHub to collaborate on open-source projects

Unit I

Typical Toolkit of DevOps : DevOps: An Overview, Achieving DevOps, Continuous Practices, How does CI Work?, Continuous Integration Practices, Commit Code Frequently, Maintain a Single Source Repository, Don't commit code, Keep the Build Fast, Every Commit Should Build the Mainline, Fix Broken Builds Immediately, Write Automated Developer Tests, Automated deployment helps, Benefits of Continuous Integration, Continuous Delivery, Continuous Delivery Process, Benefits of continuous delivery practices,

Unit II

Continuous Deployment and Version Control System : Brief about Continuous Deployment, Brief about Continuous Deployment Process, Benefits of Continuous Deployment, Version Control System (VCS), Repository and Working Copy, Types of version control systems, Benefits of Version Control Systems, Brief History of Version Control Systems, Basic Operations in a VCS, Examples of Version Control Systems, Concurrent Versions System (CVS), Subversion (SVN), Mercurial, Git, The Making - History of Linux and Git, Advantages of Git

Unit III

Centralized Version Control System vs Distributed Version Control System : Local Repository, Centralized Version Control System (CVCS), Distributed Version Control System (DVCS), Advantages of Distributed Version Control System , Private Workspace, Disadvantages of Distributed Version Control System , vs Distributed Version Control Systems, Comparison of CVCS and DVCS,

Unit IV

DVCS extended : Easier Merging, Easy to Scale Horizontally, Multiple Repositories Model, Multiple Repositories for Different Services, Resetting the Local Environment, Revert - Cancelling out the Changes

Unit V

Git : Introduction to Git, Data Structure used in Git for storing the versions, mostly used git commands with proper explanation like git add, git commit, git reset, git revert, git status, git config, git diff, git rebase, git log, git push, git pull, git fetch, git merge, git branch, git checkout, commit id, hash value generation in the commit id

Unit VI

GitHub : Introduction to GitHub, Data Structure used in GitHub to store the Data, difference between git & GitHub, types of repositories on GitHub, pull requests on GitHub, forks on GitHub, cloning a repository from GitHub

Text Books:

1. GITHUB FOR DUMMIES by PHIL HAACK SARAH GUTHALS, WILEY

References:

1. GIT POCKET GUIDE: A WORKING INTRODUCTION by RICHARD E. SILVERMAN, O'REILLY
2. EFFECTIVE DEVOPS: BUILDING A CULTURE OF COLLABORATION, AFFINITY, AND TOOLING AT SCALE by JENNIFER DAVIS, O'REILLY
3. CONTINUOUS DELIVERY: RELIABLE SOFTWARE RELEASES THROUGH BUILD, TEST, AND DEPLOYMENT AUTOMATION by JEZ HUMBLE, ADDISON-WESLEY

